10.15 The Spectral Representation: What the Exact Propagator Can and Cannot Be

LecNote11 pp. 3–7 inserts a complete set of states between two fields and, with no perturbation theory, arrives at the Lehmann–Källén form of the exact propagator: a pole at the physical mass plus a continuum from $s = 4m^2$. Students stop at four places:

- "LSZ killed the multiparticle states (LecNote4). Why are they half of the answer on p. 5?"
- "We never used the interaction. What did we learn, and where is g?"
- "$\rho \geq 0$, and $\rho = 0$ below $4m^2$. Is that all we know about ρ? Does positivity have anything to do with causality?"
- "What is the sketch on p. 7, and how do I get ρ out of a propagator I have computed?"

In short: the form of the propagator is kinematics; the dynamics is the physical mass and one non-negative function $\rho(s)$, the imaginary part of the propagator on its cut. LSZ projects onto the pole, where the continuum is finite. Positivity gives $0 \leq Z_\varphi \leq 1$ and forbids a propagator that falls faster than $1/k^2$: it is unitarity, not causality. At one loop in φ^3 theory, the divergence of $\int \rho$ is the $1/\epsilon$ that LecNote12 cancels with A. Conventions are Srednicki's: metric $(-,+,+,+)$, $k^2 = -m^2$ on shell, $\tilde{\Delta}(k) = 1/(k^2 + m^2 - i\epsilon)$, $(-\partial^2 + m^2)\Delta = \delta^d$, and $\langle 0|T\varphi(x)\varphi(y)|0\rangle = \frac{1}{i}\Delta(x-y)$ in the free theory, with $\Delta(x) = \int \frac{d^dk}{(2\pi)^d} e^{ikx}\tilde{\Delta}(k)$. Bold $\mathbf{\Delta}$ and $\tilde{\mathbf{\Delta}}$, defined in the same way from the interacting field, are the exact propagator. Spacetime has d dimensions, as on LecNote11 p. 3, $\widetilde{dk} = d^{d-1}k/\big((2\pi)^{d-1}2\omega\big)$, and the field is normalized by $\langle 0|\varphi|0\rangle = 0$ and $\langle k|\varphi(x)|0\rangle = e^{-ikx}$ (§10.6). The worked example is φ^3 theory with $\mathcal{L}_1 = \frac{1}{6}Z_g g\varphi^3 + \mathcal{L}_{\rm ct}$, $\mathcal{L}_{\rm ct} = -\frac{1}{2}A\,\partial\varphi\partial\varphi - \frac{1}{2}Bm^2\varphi^2 + Y\varphi$, $A = Z_\varphi - 1$, $B = Z_m - 1$, in $d = 6$ with $\alpha \equiv g^2/(4\pi)^3$ (LecNote12).

What Goes In, and What Comes Out

The derivation uses Poincaré invariance, with φ a scalar; a unique invariant vacuum; a complete set of states with positive norms, all with $p^0 \geq |\mathbf{p}|$ (the spectral condition); and a stable particle of mass m that φ creates, $\langle k|\varphi(0)|0\rangle \neq 0$, lighter than every multiparticle state φ reaches.

With the normalization above the result is

$$\begin{aligned}\tilde{\mathbf{\Delta}}(k^2) &= \frac{1}{k^2+m^2-i\epsilon} + \int_{4m^2}^{\infty} ds\, \frac{\rho(s)}{k^2+s-i\epsilon} \\ &= \int_0^{\infty} ds\, \frac{\hat{\rho}(s)}{k^2+s-i\epsilon}, \qquad \hat{\rho}(s) = \delta(s-m^2)+\rho(s),\end{aligned} \tag{10.15.1}$$

where, summing over the multiparticle states of total momentum k and invariant mass M_n (p. 5),

$$\rho(s) = \sum_n \left|\langle k,n|\varphi(0)|0\rangle\right|^2 \delta(s-M_n^2) \;\geq 0. \tag{10.15.2}$$

Everything in (10.15.1) except m and ρ is kinematics; computing them takes perturbation theory (LecNote12) or a non-perturbative method. Conversely, a propagator computed order by order must take this form, pole at the physical m with residue 1, and demanding that fixes A and B: $\Pi(-m^2)=\Pi'(-m^2)=0$ (LecNote12 p. 3; §10.11). No step uses the size of g, so (10.15.1) holds at strong coupling (p. 7), as long as the stable particle exists. If φ creates no single particle, there is no pole term; if the particle can decay, it is not an eigenstate of H, and the spike becomes a peak inside the continuum (below (10.15.5)).

Why ρ depends on s only, and where it starts

Between pp. 5 and 6, insert $1=\int ds\,\delta(s-M_n^2)$ in each term. Since $\varphi(0)$, $|0\rangle$ and the measure are Lorentz invariant ($d^{d-1}k/\omega$ is boost invariant), the combination $\sum_n |\langle k,n|\varphi(0)|0\rangle|^2\delta(s-M_n^2)$ can depend on k only through $k^2=-s$. It is a function of s alone and leaves the $\widetilde{dk}$ integral, and each s contributes a free Wightman function of mass^2 s:

$$\begin{aligned}\langle 0|\varphi(x)\varphi(y)|0\rangle &= \int_0^{\infty} ds\,\hat{\rho}(s)\,D_s(x-y), \\ D_s(x) &= \int \frac{d^{d-1}k}{(2\pi)^{d-1}\,2\omega_s}\,e^{ikx}, \qquad k^0=\omega_s=(\mathbf{k}^2+s)^{1/2}.\end{aligned} \tag{10.15.3}$$

The θ's of the time ordering (p. 6) carry no s, so the one-mode identity of §10.8 at mass^2 s turns each term into $1/(k^2+s-i\epsilon)$. Nobody adds the $-i\epsilon$: it is the time ordering.

The continuum starts at the smallest invariant mass of the multiparticle states $\varphi(0)$ reaches. In φ^3 theory φ couples to two particles already at order g, and the lightest pair has both at rest, $M=2m$: hence $4m^2$. If the theory and its vacuum are symmetric under $\varphi\to-\varphi$ (pure φ^4, say), n particles have charge $(-1)^n$ and φ is odd, so $\langle n|\varphi(0)|0\rangle=0$ unless n is odd, and the continuum starts at three particles at rest, $s=9m^2$. By the same sum, a bound state of mass $m_B<2m$ that φ creates adds a spike $Z_B\,\delta(s-m_B^2)$ to $\hat{\rho}$, with $Z_B=|\langle k,B|\varphi(0)|0\rangle|^2$, and a second pole $Z_B/(k^2+m_B^2-i\epsilon)$ to the propagator.

Pole, cut and imaginary part

As a function of complex k^2, (10.15.1) is analytic except for a pole at $k^2 = -m^2$ with residue 1 and a branch cut along $k^2 \le -4m^2$. On the real axis the $-i\epsilon$ picks the physical side of the cut, and $1/(x - i\epsilon) = P(1/x) + i\pi\delta(x)$ gives

$$\operatorname{Im} \tilde{\Delta}(-s) = \pi\hat{\rho}(s) = \pi\big[\delta(s - m^2) + \rho(s)\big]. \tag{10.15.4}$$

At $k^2 + i\epsilon$ the sign flips: the jump across the cut is $2\pi i\rho(s)$. So to read ρ off a propagator you have computed, take its imaginary part at $k^2 = -s$; read the other way, (10.15.1) is a dispersion relation that rebuilds $\tilde{\Delta}$ from it. The sketch on p. 7 is $\hat{\rho}$ against the mass $\sqrt{s}$: the spike at m is the pole term, the continuum starts at $2m$. Fig. 10.4 shows the one-loop φ^3 result worked out below.

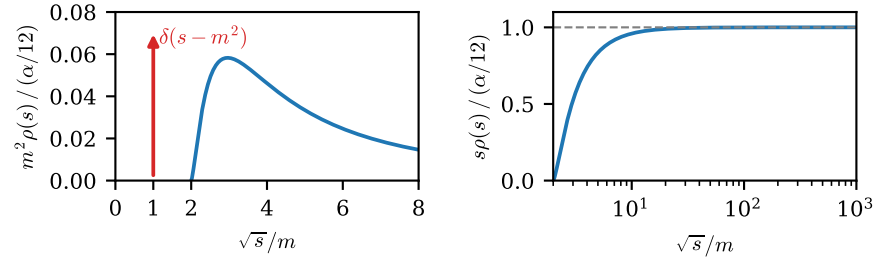


Figure 10.4: One-loop spectral weight of φ^3 theory in $d = 6$, Eq. (10.15.12). Left: $\hat{\rho}$ against $\sqrt{s}$, the computed version of the sketch on LecNote11 p. 7; the arrow is the pole term $\delta(s - m^2)$, of weight 1 (not to scale). Right: $s\rho(s) \to \alpha/12$, so $\int \rho\, ds$ grows like $\frac{\alpha}{12} \ln s$.

With $\tilde{\Delta} = 1/(k^2 + m^2 - i\epsilon - \Pi(k^2))$ (LecNote12 p. 2), (10.15.4) reads, for $s \ge 4m^2$,

$$\pi\rho(s) = \frac{\operatorname{Im}\Pi(-s)}{\big[m^2 - s - \operatorname{Re}\Pi(-s)\big]^2 + \big[\operatorname{Im}\Pi(-s)\big]^2}, \tag{10.15.5}$$

the formula at the end of LecNote12 (p. 15). The denominator is positive, so $\rho \ge 0$ is $\operatorname{Im}\Pi \ge 0$ on the cut; to lowest order $\rho = \operatorname{Im}\Pi(-s)/\pi(s - m^2)^2$. If the particle can decay, $\operatorname{Im}\Pi > 0$ already where $m^2 - s - \operatorname{Re}\Pi$ vanishes, so the denominator never reaches zero: instead of a spike, $\pi\hat{\rho}$ has a peak of height $1/\operatorname{Im}\Pi$ there (m is then a mass parameter, not a pole).

LSZ and the continuum

LecNote4 showed that the multiparticle states drop out of LSZ (the red box on p. 5 recalls it), yet they are half of (10.15.1). The two statements are about different objects. What decays in LSZ is the overlap of a multiparticle state with the packet operator, $\langle s|a_1^\dagger(t)|0\rangle \propto e^{iK_s t}$ with

$K = H - (\mathbf{P}^2 + m^2)^{1/2}$, Eqs. (10.6.2)–(10.6.3): K vanishes on the one-particle shell and varies over the continuum, so the multiparticle part dephases as $t \to \pm\infty$, a weak limit, not a vanishing amplitude. Here nothing is smeared or sent to $t \to \pm\infty$: $\langle k, n|\varphi(x)|0\rangle = e^{-ikx}\langle k, n|\varphi(0)|0\rangle$ has the same modulus at every x, and every state φ couples to enters ρ. The dephasing shows in (10.15.3) too: at fixed k the pole term oscillates forever, while the continuum, a spread of frequencies $(\mathbf{k}^2 + s)^{1/2}$, dies out (for the one-loop ρ below, as $t^{-5/2}$ at $\mathbf{k} = 0$).

In momentum space the LSZ step multiplies by $k^2 + m^2$ and goes on shell:

$$(k^2 + m^2)\,\tilde{\mathbf{\Delta}}(k^2) = 1 + (k^2 + m^2)\int_{4m^2}^{\infty} ds\,\frac{\rho(s)}{k^2 + s - i\epsilon} \;\to\; 1. \tag{10.15.6}$$

At $k^2 = -m^2$, $k^2 + s \geq 3m^2$ on the whole continuum, so the integral is finite and its prefactor kills it. The continuum never left the correlators; LSZ projects onto the pole.

The sum rule, and what positivity forbids

Subtract from (10.15.3) the same expression with $x \leftrightarrow y$:

$$\langle 0|[\varphi(x), \varphi(y)]|0\rangle = \int_0^{\infty} ds\,\hat{\rho}(s)\,C_s(x - y), \tag{10.15.7}$$

where $C_s(x) = D_s(x) - D_s(-x)$ is the commutator of a free field of mass^2 s, a c-number. At equal times $\partial_{y^0} C_s(x - y) = i\delta^{d-1}(\mathbf{x} - \mathbf{y})$ for every s: mode by mode, ∂_{y^0} brings down $\pm i\omega_s$, which cancels the $1/2\omega_s$. So $\langle 0|[\varphi(\mathbf{x}, t), \partial_0\varphi(\mathbf{y}, t)]|0\rangle = i\delta^{d-1}(\mathbf{x} - \mathbf{y})\,[1 + \int ds\,\rho(s)]$. The kinetic term of $\mathcal{L}_0 + \mathcal{L}_{\rm ct}$ is $-\frac{1}{2}Z_\varphi\partial\varphi\partial\varphi$, so the canonical momentum of the field normalized by $\langle k|\varphi(0)|0\rangle = 1$ is $Z_\varphi\partial_0\varphi$ (see below (10.6.21)), and the commutator is $i\delta^{d-1}/Z_\varphi$. Hence

$$1 + \int_{4m^2}^{\infty} ds\,\rho(s) = \frac{1}{Z_\varphi}. \tag{10.15.8}$$

With $\rho \geq 0$ this gives $0 \leq Z_\varphi \leq 1$, with $Z_\varphi = 1$ only if $\varphi(0)|0\rangle$ has no multiparticle component.

For real $k^2 > -m^2$ every term of (10.15.1) is positive, so

$$\tilde{\mathbf{\Delta}}(k^2) \geq \frac{1}{k^2 + m^2} \qquad (k^2 > -m^2). \tag{10.15.9}$$

The exact propagator cannot fall faster than $1/k^2$ at large spacelike k^2; if $\int\rho$ converges, $k^2\tilde{\mathbf{\Delta}}$ rises toward $1/Z_\varphi$. A regulator that makes the propagator fall faster, so that loops converge, must give up positive norms. Pauli–Villars (LecNote12 p. 8) does:

$$\frac{1}{p^2 + m^2}\,\frac{\Lambda^2}{p^2 + \Lambda^2} = \frac{\Lambda^2}{\Lambda^2 - m^2}\Big[\frac{1}{p^2 + m^2} - \frac{1}{p^2 + \Lambda^2}\Big]. \tag{10.15.10}$$

This has the form of (10.15.1) with a state at $s = \Lambda^2$ of weight $-\Lambda^2/(\Lambda^2 - m^2) < 0$: a negative-norm ghost. The weights add to zero, as a $1/p^4$ falloff requires, and the residue at m^2 is $\Lambda^2/(\Lambda^2 - m^2)$, not 1; both defects go away only as $\Lambda \to \infty$.

Causality is not positivity

Each C_s in (10.15.7) vanishes at spacelike separation. For spacelike x there is a frame with $x^0 = 0$, where $D_s(0, \mathbf{x}) = \int \widetilde{dk}\, e^{i\mathbf{k}\cdot\mathbf{x}}$ is even in $\mathbf{x}$; D_s is Lorentz invariant, so $D_s(x) = D_s(-x)$ in every frame. Hence $\langle 0|[\varphi(x), \varphi(y)]|0\rangle = 0$ for spacelike $x - y$ whatever the real weights $\hat{\rho}$: a Pauli–Villars pair $\pm\Lambda^2/(\Lambda^2 - m^2)$ is just as causal. The argument used Lorentz invariance and the spectral condition (only $s \geq 0$ occurs, so each D_s lives on a forward mass shell that Lorentz transformations map onto itself), never the sign of ρ. (This is the vacuum value; that the operators themselves commute at spacelike separation is locality, an input of the theory.)

What $\rho \geq 0$ buys is probability. By (10.15.2), ρ is a sum of squared matrix elements; a negative ρ needs a state of negative norm, hence negative probabilities: unitarity fails. The Pauli–Villars ghost breaks unitarity, not causality. Everything above that used the sign of ρ is unitarity at work: $Z_\varphi \leq 1$, the bound (10.15.9), and $\operatorname{Im}\Pi \geq 0$ on the cut.

A worked check: φ^3 in six dimensions at one loop

At order g^2 only two-particle states contribute to (10.15.2). With a factor $\frac{1}{2}$ for identical particles in their completeness relation, and with $d^dk/(2\pi)^d = (ds/2\pi)\,\widetilde{dk}$ at $k^0 = \omega_s$,

$$\rho(s) = \frac{1}{4\pi} \int \widetilde{dk_1}\, \widetilde{dk_2}\, (2\pi)^d \delta^d(k - k_1 - k_2)\, \big|\langle k_1 k_2|\varphi(0)|0\rangle\big|^2. \tag{10.15.11}$$

To lowest order, $\langle k_1 k_2|\cdots|0\rangle$ of the field equation $(-\partial^2 + m^2)\varphi = \frac{1}{2}g\varphi^2 + \ldots$ gives $(m^2 - s)\langle k_1 k_2|\varphi(0)|0\rangle = g$. In $d = 6$ the two-body phase space is $s\beta^3/(192\pi^2)$, with $\beta = (1 - 4m^2/s)^{1/2}$. Hence

$$\rho(s) = \frac{\alpha}{12}\, \frac{s\,\beta^3}{(s - m^2)^2}\, \theta(s - 4m^2), \qquad \alpha = \frac{g^2}{(4\pi)^3}. \tag{10.15.12}$$

Independently, LecNote12 p. 13 has $\Pi = \frac{1}{2}\alpha \int_0^1 dx\, D \ln(D/D_0) - \frac{\alpha}{12}(k^2 + m^2)$, with $D = m^2 + x(1 - x)k^2$; the $-i\epsilon$ of the propagators makes it $D - i\epsilon$. At $k^2 = -s$, $D < 0$ for $x_- < x < x_+$, $x_\pm = \frac{1}{2}(1 \pm \beta)$, which needs $s > 4m^2$; there $\operatorname{Im}(D \ln D) = \pi[x(1 - x)s - m^2]$, and integrating gives $\operatorname{Im}\Pi(-s) = \frac{\pi\alpha}{12} s\beta^3$. Then $\operatorname{Im}\Pi/\pi(s - m^2)^2$ is (10.15.12) exactly. The one-loop ρ is non-negative and

vanishes below $4m^2$, as p. 5 says. It rises like $(s-4m^2)^{3/2}$, the p^{d-3} of two-body phase space at $d=6$, peaks near $\sqrt{s}\approx 3m$, and falls like $\alpha/(12s)$ (Fig. 10.4).

Dispersion relation. At this order $\tilde{\mathbf{\Delta}} = 1/(k^2+m^2) + \Pi/(k^2+m^2)^2$, so (10.15.1) claims

$$\Pi(k^2) = (k^2+m^2)^2 \int_{4m^2}^{\infty} ds\, \frac{\rho(s)}{k^2+s} + O(\alpha^2), \tag{10.15.13}$$

a dispersion relation with two subtractions at $k^2=-m^2$, which are $\Pi(-m^2)=\Pi'(-m^2)=0$. Evaluated numerically, it reproduces LecNote12's on-shell Π to ten digits at $k^2/m^2 = -2,\ 0,\ 1,\ 10,\ 100$. A calculation that never mentioned a counterterm returns the renormalized self-energy.

Sum rule. Since $s\rho \to \alpha/12$, $\int \rho\, ds$ diverges logarithmically, and (10.15.8) gives a regulator-dependent $Z_\varphi = 1 - \int \rho\, ds + O(\alpha^2)$. Use LecNote12's regulator: in $d=6-\epsilon$, with $g \to g\tilde{\mu}^{\epsilon/2}$, the same phase-space calculation gives $\rho \propto s^{-1-\epsilon/2}$ at large s, the integral converges, and its pole is $\frac{\alpha}{12}\cdot\frac{2}{\epsilon}$:

$$Z_\varphi - 1 = -\int ds\, \rho(s) = -\frac{\alpha}{6\epsilon} + O(\epsilon^0). \tag{10.15.14}$$

This is the pole of $A = -\frac{\alpha}{6}\left[\frac{1}{\epsilon} + \ln\frac{\mu}{m} + \frac{1}{2} + \kappa_A\right]$ (LecNote12 p. 12), sign included. Not only the pole: at each ϵ, $-\int \rho\, ds$ equals the whole on-shell A fixed by $\Pi'(-m^2)=0$ (to 30 digits for ϵ from 1 to 0.002), as (10.15.8) requires. With a cutoff on s instead, $\int_{4m^2}^{\Lambda^2} \rho\, ds = \frac{\alpha}{12}[\ln(\Lambda^2/m^2) + 3 - \pi\sqrt{3}] + O(m^2/\Lambda^2)$: $\ln\Lambda^2 \leftrightarrow 2/\epsilon$, as for a cutoff on the Euclidean loop momentum (§10.18 shows it in $d=4$). The same constant returns at large spacelike k^2, where $k^2\tilde{\mathbf{\Delta}} - 1 \simeq \frac{\alpha}{12}[\ln(k^2/m^2) + 3 - \pi\sqrt{3}]$, the logarithm of LecNote12 p. 15: the propagator at k^2 has counted the spectral weight up to $s\approx k^2$, and the divergent sum rule is the slow growth of $k^2\tilde{\mathbf{\Delta}}$.

Summary

- (10.15.1) is kinematics: a pole at the physical mass with residue 1 and a continuum $\rho \geq 0$ from the lowest threshold ($4m^2$ in φ^3, $9m^2$ with $\varphi\to-\varphi$), plus a spike per bound state. The dynamics is m and $\rho = \operatorname{Im}\tilde{\mathbf{\Delta}}(-s)/\pi$.
- LSZ projects onto the pole, where the continuum is finite. The sum rule $1 + \int\rho = 1/Z_\varphi$ and positivity give $0 \leq Z_\varphi \leq 1$ and $\tilde{\mathbf{\Delta}} \geq 1/(k^2+m^2)$ for $k^2 > -m^2$. The vacuum commutator vanishes outside the light cone for any sign of ρ: positivity is unitarity, not causality.
- One loop, $d=6$: $\rho = \frac{\alpha}{12} s\beta^3/(s-m^2)^2$ rebuilds the on-shell Π; the divergence of $\int\rho$ is the $1/\epsilon$ of A.